Data Science and Visualization

Unit II

DESCRIPTIVE STATISTICS

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Learning Augmented

Unit II – Descriptive Statistics

Advanced Attribute Types - Measures of Central Tendency - Measures of Dispersion - Skewness and Kurtosis - Visual Outlier Identification - Missing Data Visualization - Exploratory Data Analysis (EDA) Techniques - Histograms, KDEs, Box Plots, and Violin Plots.

CO2	**Perform Exploratory Data Analysis by applying descriptive statistics and univariate/bivariate visualizations to uncover data distributions, patterns, and relationships, while analyzing data quality through the detection of outliers and exploration of missing value patterns.**

Unit II – Descriptive Statistics

Deep Dive into Data/Attribute Types: Nominal-Ordinal-Binary-Numeric -Discrete vs. Continuous-Asymmetric Attributes-Measurement Scales - Basic Statistical Descriptions: Measures of Central Tendency -Mean-Median-Mode - Measures of Dispersion -Range-Quartiles-Variance-Standard Deviation-IQR - Understanding Skewness and Kurtosis -Conceptual - Data Cleaning Context: Visual identification of outliers - Visualizing Missing Data –Types of Missingness- Exploratory Data Analysis Visualization Techniques-Univariate: Histograms with Kernal Density Estimation-Box Plots -Violin Plots.

CO2	**Perform Exploratory Data Analysis by applying descriptive statistics and univariate/bivariate visualizations to uncover data distributions, patterns, and relationships, while analyzing data quality through the detection of outliers and exploration of missing value patterns.**

What is Data?

- Datasets are made up of **data objects**.
- A **data object** represents an entity.
  - Also called *sample, example, instance, data point, object, tuple.*
- **Attributes describe data objects.**
- An **attribute** is a property or characteristic of a data object.
  - **Examples:** Eye colour, temperature, etc.
- Attribute is also known as a *variable, field, characteristic or feature.*
- A collection of attributes describes an object.

Attributes

Objects

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

What are Attributes?

- Columns of the dataset are called Attributes, which represent the characteristics of the data object.
- Rows of a dataset are called **Data Objects**.
- **Other Names**
  - ✓ **Attributes**- Data Scientist/Database Professionals
  - ✓ **Dimension**-Data Warehousing
  - ✓ **Feature**-Machine Learning
  - ✓ **Variables**-Statisticians

Attributes

Objects

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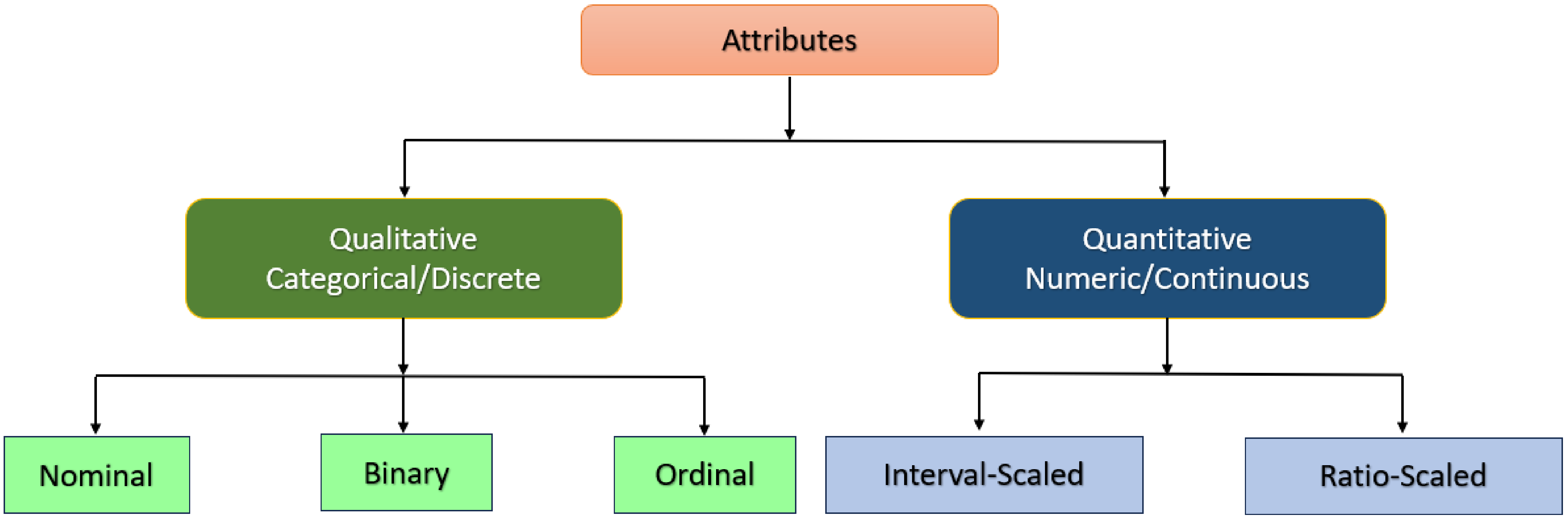
Attribute Types



Nominal Attributes

Qualitative/Categorical/Discrete

- The values of a nominal attribute are **symbols or names of things**.
- Each value represents **some kind of category, code, or state.**
- The values of nominal attributes **do not have any meaningful order.**

Example

Attributes	Possible Values
hair_color	black, brown, red, green, and so on.
marital_status	single, married, divorced, and widowed.
occupation	teacher, doctor, farmer, student and so on.

Binary Attributes
Qualitative/Categorical/Discrete

- A special nominal attribute with **only two states 0 or 1.**

Symmetric Binary Attribute	Asymmetric Binary Attribute
Both states are **equally valuable** and carry the same weight	the outcomes of the states are **not equally important**
Ex. Gender- Both male and female are important	Ex.Test results for COVID patient: Positive (1) and Negative (0)

Ordinal Attributes

Qualitative/Categorical/Discrete

- Values that have **meaningful order or ranking** *but magnitude between successive values is not known.*

Examples	
Socio economic status	Low income, middle income, high income
Educational Level	High school, BS, MS, PhD
Income level	Less than 50K,50-100K,over 100K
Customer satisfaction	0: very dissatisfied 1: somewhat dissatisfied 2: neutral 3: satisfied 4: very satisfied

Interval-Scaled Attributes

Quantitative/Numeric/Continuous

- Measured on a scale of **equal-sized units.**
- It is a measurable quantity represented in **integer or real values.**
- Values can be **ordered** and can be positive, 0 or negative.
- Can **quantify the difference** between values.
- **Examples:**
  - ✓Temperature: Measured in Fahrenheit or Celsius
  - ✓Credit Scores: Measured from 300 to 850
  - ✓SAT Scores: Measured from 400 to 1,600

Ratio-Scaled Attributes

Quantitative/Numeric/Continuous

- It is a scale used to label variables that have a **natural order, a quantifiable difference** between values, and a true zero value.

Examples:

✓Height: His height is twice of that person

✓Weight

✓Temperature

Level of Measurement

Nominal	Ordinal	Interval	Ratio
"Eye color"	"Level of satisfaction"	"Temperature"	"Height"
Named	Named	Named	Named
	Natural order	Natural order	Natural order
		Equal interval between variables	Equal interval between variables
			Has a "true zero" value, thus ratio between values can be calculated

Level of Measurement

Property	Nominal	Ordinal	Interval	Ratio
Has a natural "order"	NO	YES	YES	YES
Mode can be calculated	YES	YES	YES	YES
Median can be calculated		YES	YES	YES
Mean can be calculated			YES	YES
Exact difference between values			YES	YES
Has a "true zero" value				YES

Discrete Vs. Continuous Attributes

Discrete Attribute

- Has only a finite or countably infinite set of values.
- E.g., zip codes, profession, or the set of words in a collection of documents.

Continuous Attribute

- Has real numbers as attribute values.
- E.g., temperature, height, or weight.
- Continuous attributes are typically represented as floating-point variables.

Statistics

Statistics - Definition

- Science and branch of mathematics dedicated to **collecting, organizing, analyzing, interpreting, and presenting data** to turn raw information into meaningful insights.

- A **data scientist** is a person who is *better at statistics* than any programmer and *better at programming* than any statistician.

Functions of Statistics

- It presents fact
- It helps in prediction
- It helps in test hypothesis
- It facilitate comparison

Types of Analysis in Statistics

Quantitative Analysis

- Collecting and interpreting the data with numbers and graphs to identify patterns and trends.

Qualitative Analysis

- Gives generic information and uses text, sound and other forms of media to do so.

Types of Analysis in Statistics

Quantitative Analysis

- The shop sells 70 shirts in a week.

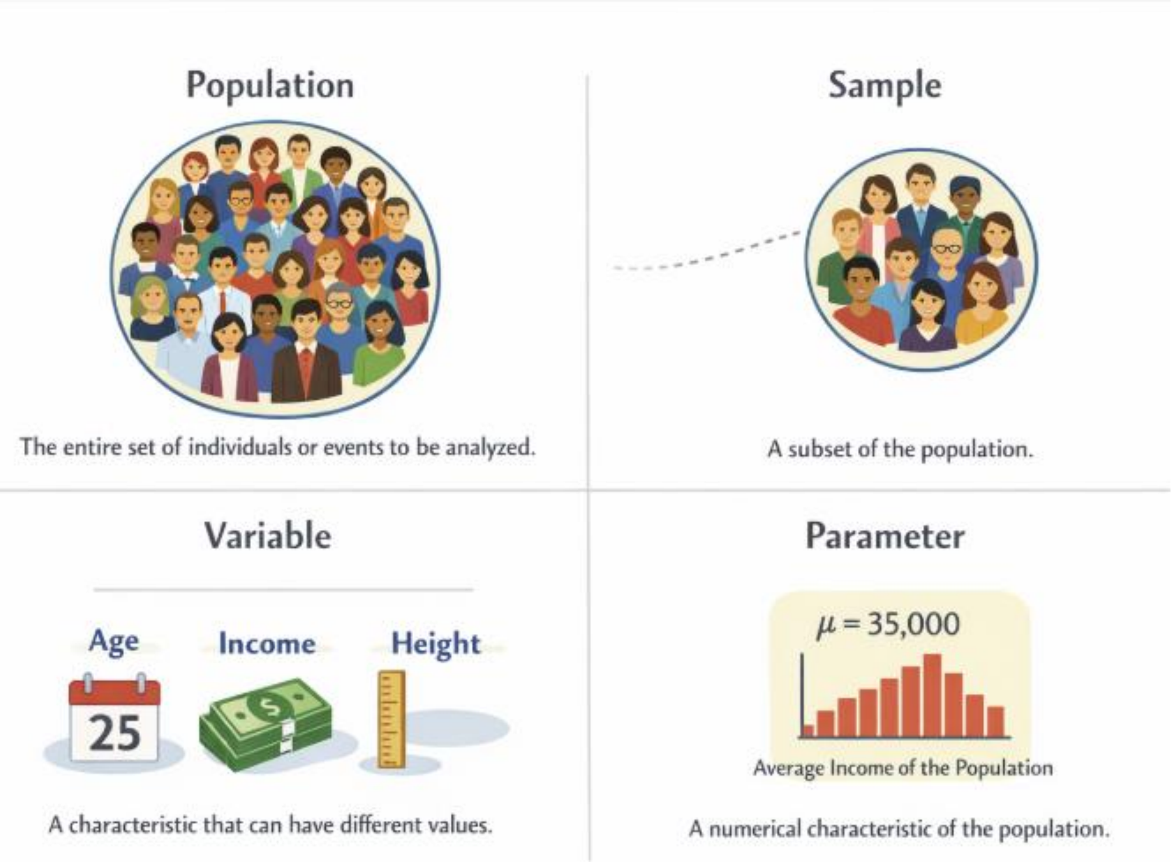
Qualitative Analysis

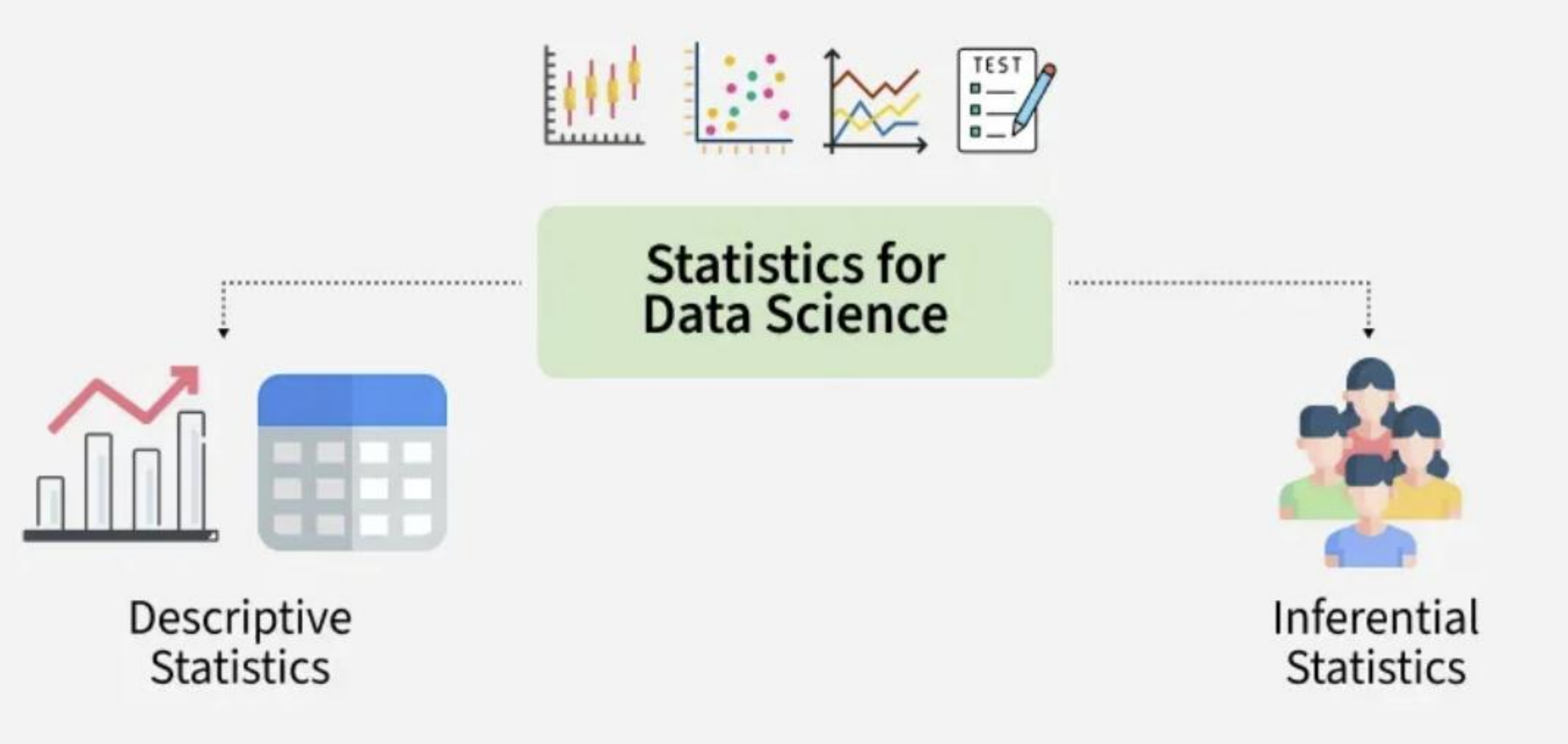
- The Shirt is available in the sizes Small, Medium, Large.

Statistics Terminologies

Some of the most common terms you might come across in statistics are:

- **Population:** It is a collection of a set of individual objects or events whose properties are to be analyzed.
- **Sample:** It is the subset of a population.
- **Variable:** It is a characteristic that can have different values.
- **Parameter:** It is numerical characteristic of population.

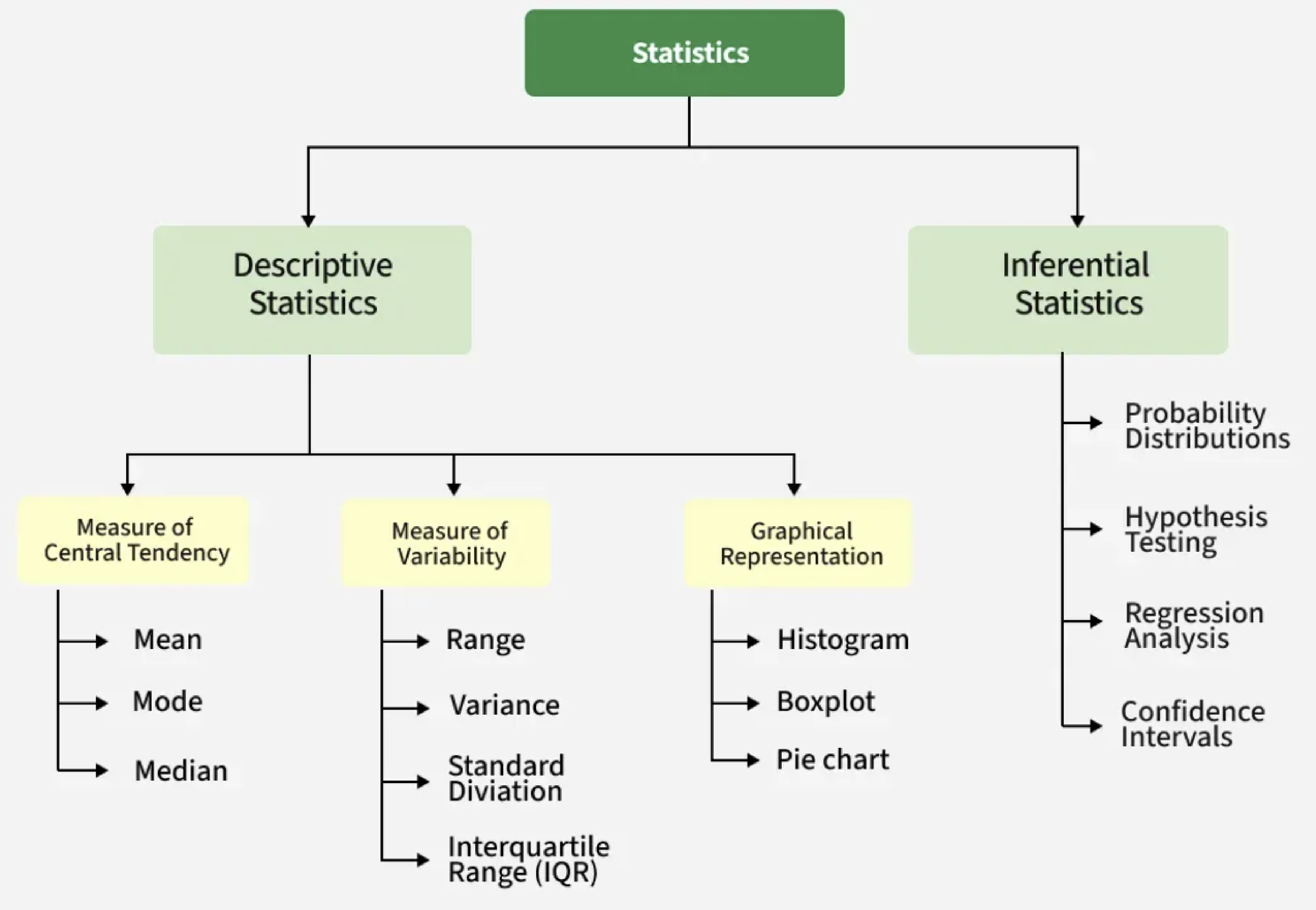


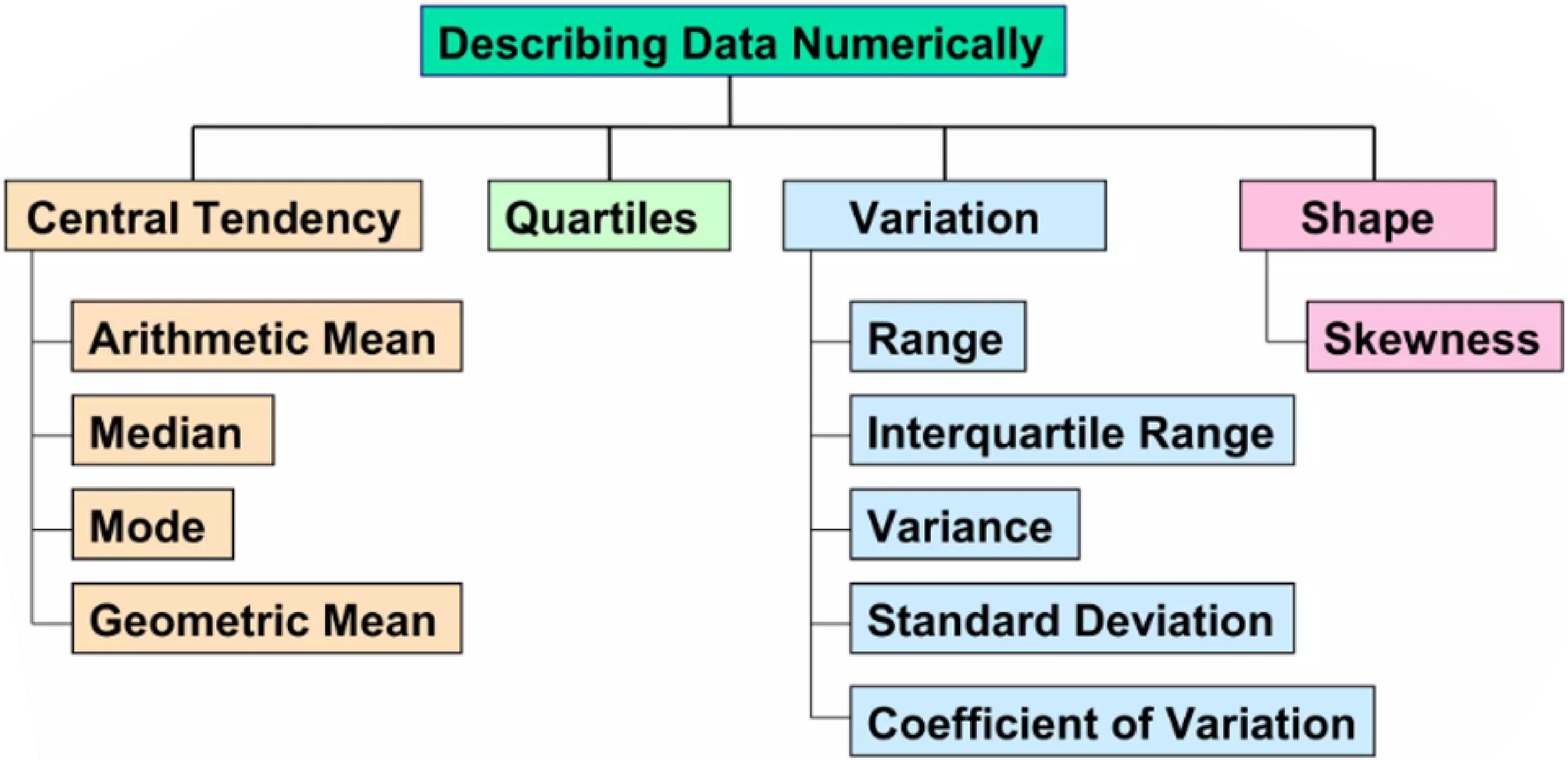


TEST
Statistics for Data Science
Descriptive Statistics
Inferential Statistics

Categories in Statistics

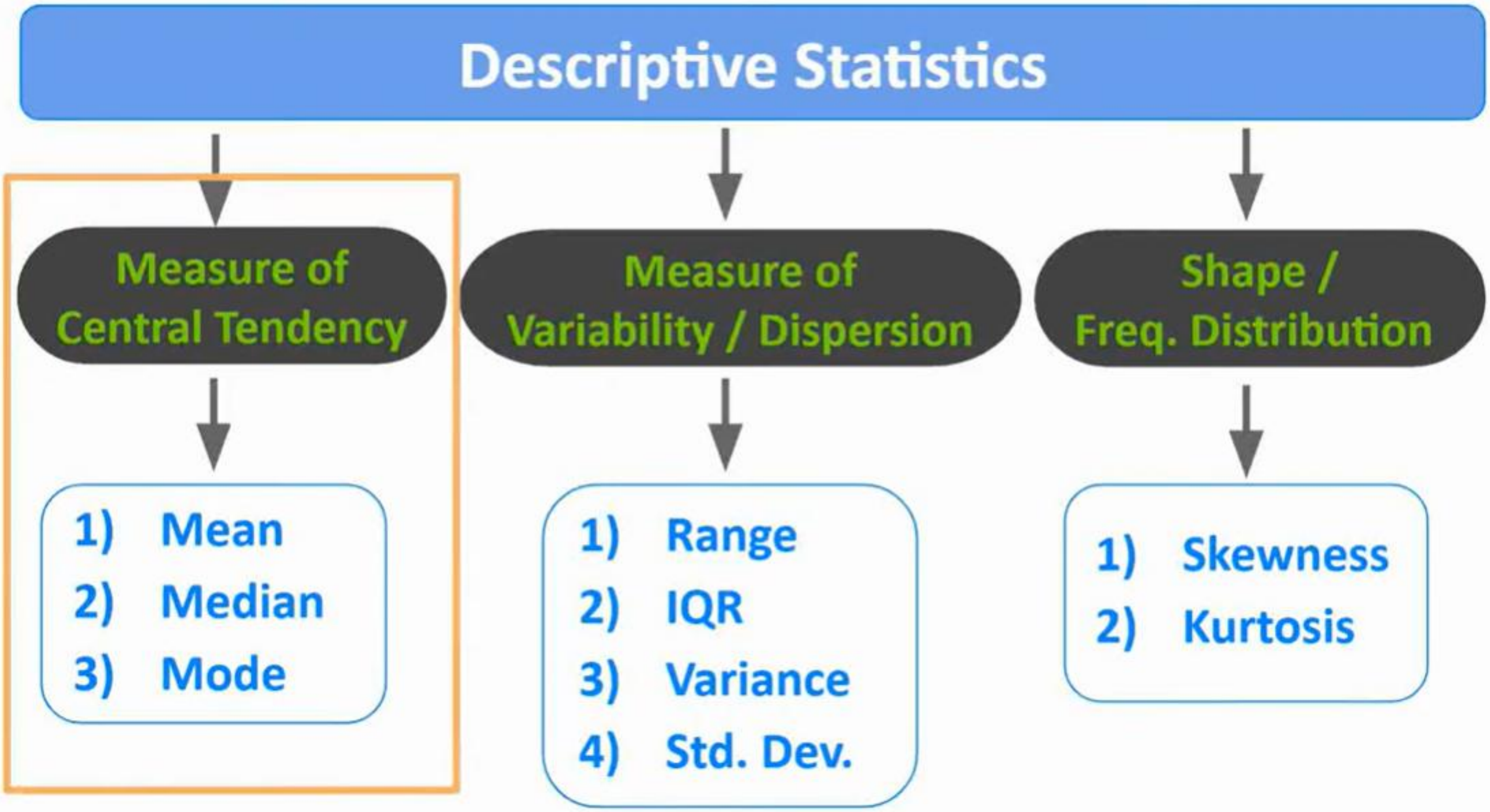
Descriptive Statistics	Inferential Statistics
It uses the data to provide descriptions about the population.	It makes predictions about a population based on a sample of data taken from the population.
E.g., Charts, Tables, Infographics.	E.g., T-Test, ANOVA Test.
Tools-Data Distribution, Skewness, Correlation, Central Tendency.	**Tools-**Hypothesis Testing, regression Analysis


Statistics
Descriptive Statistics
Inferential Statistics
Measure of Central Tendency
Mean
Mode
Median
Measure of Variability
Range
Variance
Standard Diviation
Interquartile Range (IQR)
Graphical Representation
Histogram
Boxplot
Pie chart
Probability Distributions
Hypothesis Testing
Regression Analysis
Confidence Intervals


Describing Data Numerically
Central Tendency
Arithmetic Mean
Median
Mode
Geometric Mean
Quartiles
Variation
Range
Interquartile Range
Variance
Standard Deviation
Coefficient of Variation
Shape
Skewness

Basic Statistical Descriptions of Data

- Identify properties of the data and **highlight noise or outliers**.
- Measures of central tendency include **mean, median, mode and midrange.**
- Measures of data dispersion include **quartiles, interquartile range (IQR) and variance.**
- These descriptive statistics are of great help in understanding the **distribution of the data.**


Descriptive Statistics
Measure of Central Tendency
Measure of Variability / Dispersion
Shape / Freq. Distribution
1) Mean
2) Median
3) Mode
1) Range
2) IQR
3) Variance
4) Std. Dev.
1) Skewness
2) Kurtosis

Descriptive Statistics

- Descriptive statistics uses data that describes the population either through numerical calculations, graphs, or tables.
- It provides a **graphical summary of data**.
- It is simply used for **summarizing objects**, etc.
- There are two categories in this, as follows.
  - ✓**Measure of Central Tendency**
  - ✓**Measure of Variability**

Descriptive Statistics

- Descriptive statistics uses data that **describes the population either through numerical calculations, graphs, or tables.**
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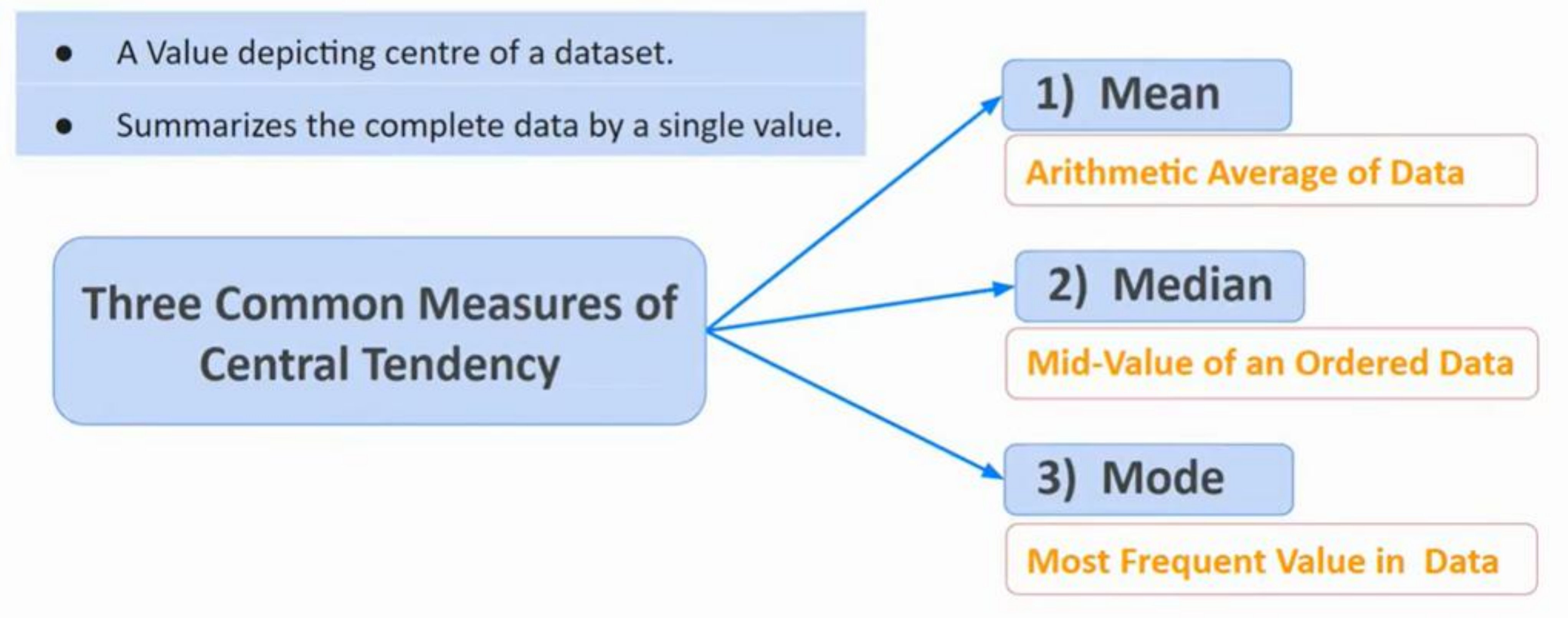
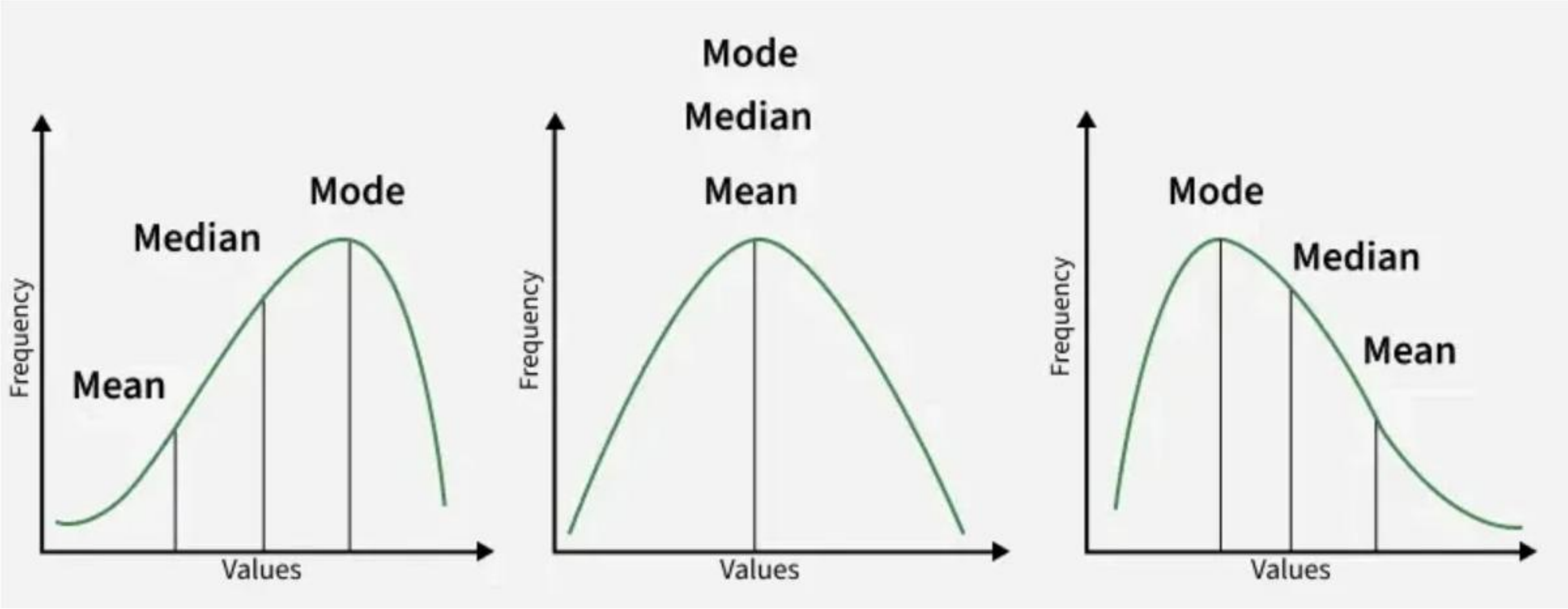
  - ✓**Measure of Central Tendency**
  - ✓**Measure of Variability**

Measure of Central Tendency

Measure of central tendency is also known as a **summary statistic** that is used to represent the **center point or a particular value of a data set or sample set.**



Measure of Central Tendency



Measure of Central Tendency - MEAN

Formula

$$\text{Mean} = \frac{\text{Sum (all datapoints)}}{\text{Number datapoints}}$$

Example: Mean Income

$$£2{,}036.50 = \frac{1250 + 2760 + 1880 + 2560 + 1750 + 2345 + 1905 + 1905 + 1985 + 2205}{10}$$

Person 1: Income: £1,250

Person 2: Income: £2,760

Person 3: Income: £1,880

Person 4: Income: £2,560

Person 5: Income: £1,750

Person 6: Income: £2,345

Person 7: Income: £1,905

Person 8: Income: £1,905

Person 9: Income: £1,985

Person 10: Income: £2,205

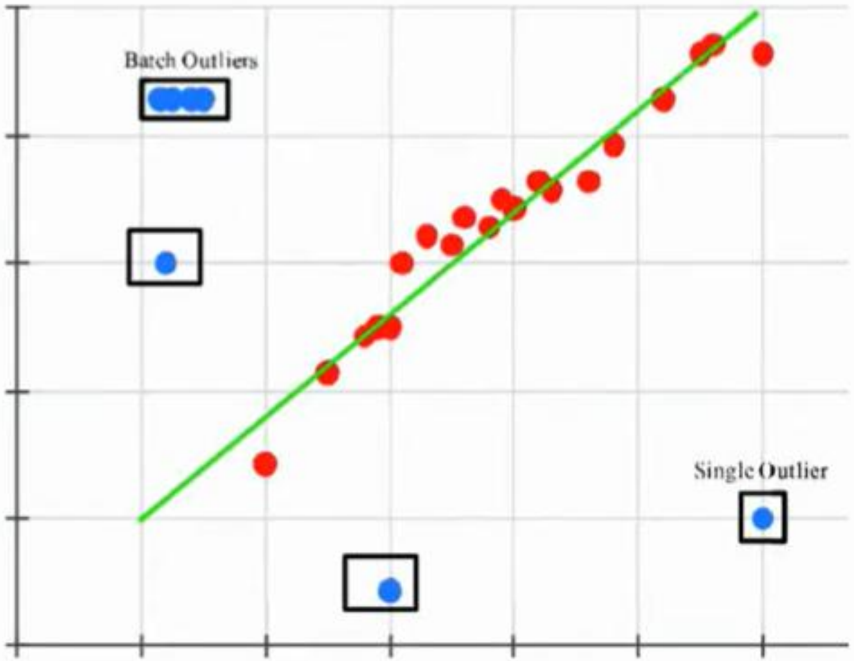
1. MEAN

- Most Stable Measure, **WHY ?** → *because it includes all the values as a part of the calculation.*
- But Susceptible to Outliers.

Eg. Employee Salary

4k, 5k, 1k, 3k, 7k	*Mean* →	20/5 = 4.0 ✔
4k, 5k, 1k, 3k, 7k, 98k	*Mean* →	128/6 = 19.6

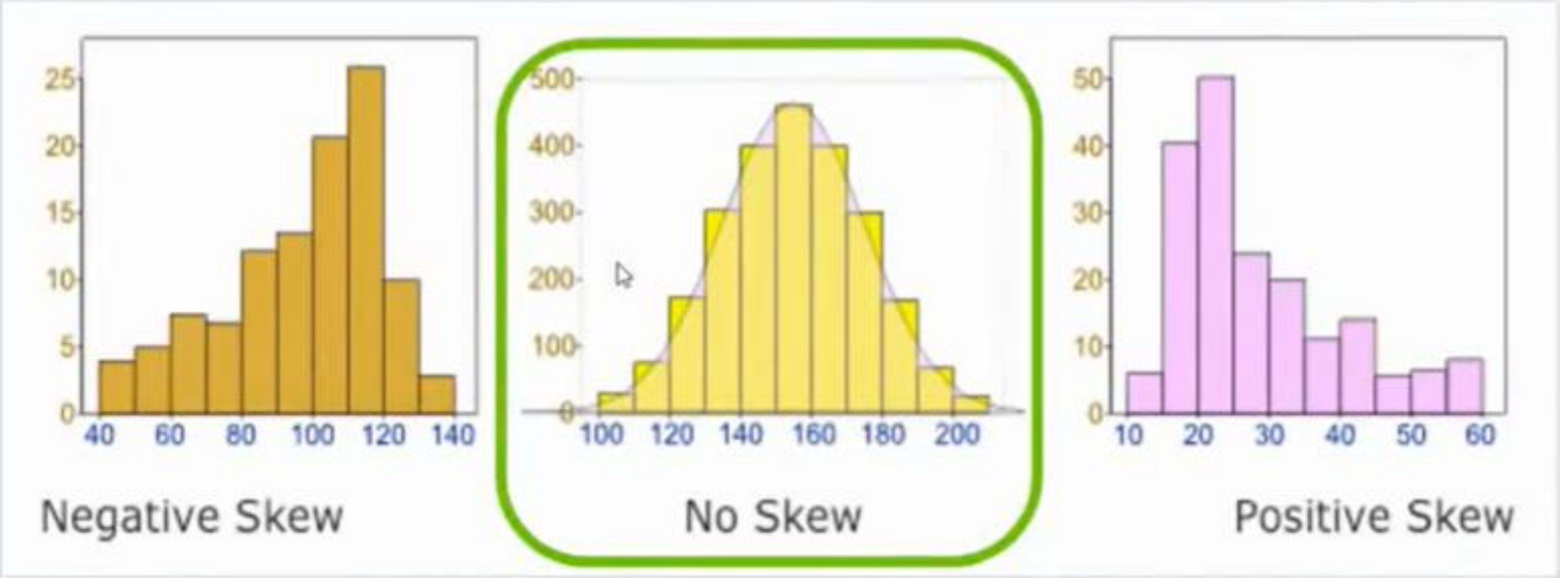
An Outlier — *Unexpected Increase in Mean*



NOTE *: A single outlier can have a great effect on the Mean.*

WHEN TO USE MEAN ?

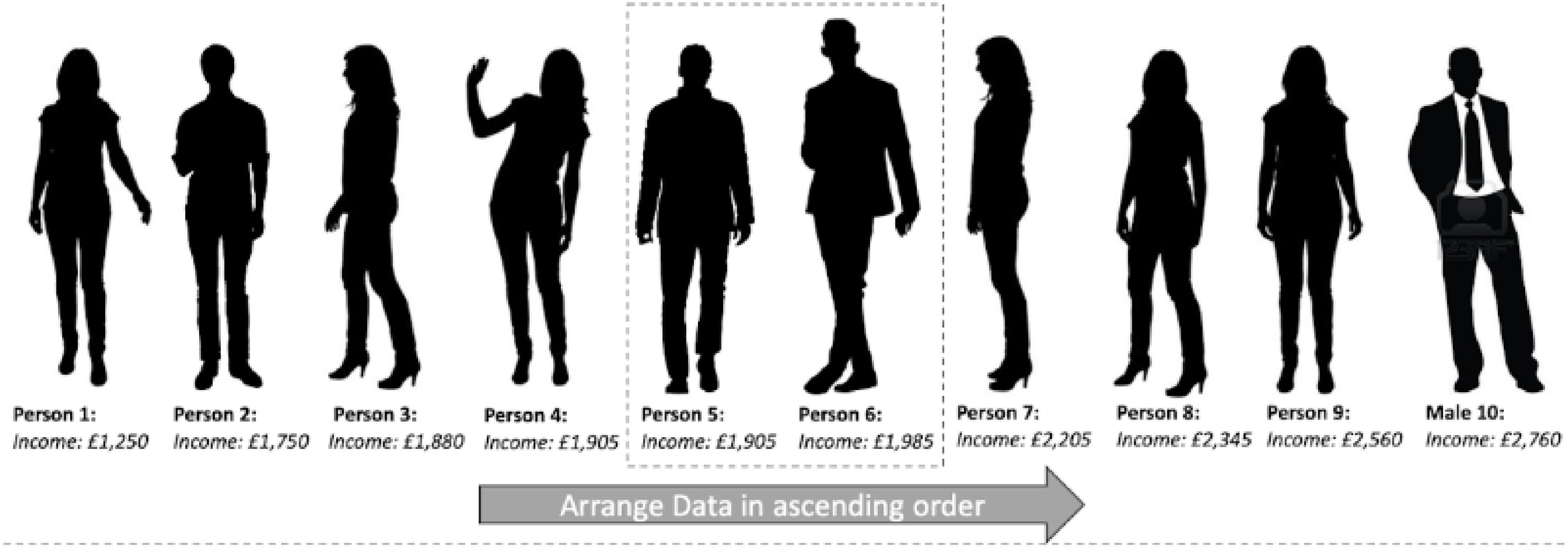
- If Data consists NO Significant **Outliers / Extreme values.**
- If the Data Distribution is NOT **Skewed.**



Measure of Central Tendency - MEDIAN

Median = Middle value (if odd) or mean of the two middle values (if even)

$$£1,945 = \frac{£1925 + £\ 1985}{2}$$



2. MEDIAN

- Middle value of an ordered dataset.
- Not Susceptible to Outliers.

To Compute Median : Firstly, Sort the values to have an Ordered data.

If No. of values is **EVEN** : ★ Median = Avg. of two mid-values	Eg. 2, 4, 6, 8, 10, 98	= (6+8)/2 = 7
If No. of values is **ODD** : ★ Median = Middle value	Eg. 2, 4, 6, 7, 98	= 6

WHEN TO USE MEDIAN ?

- If Data consists some **Outliers / Extreme values.** (A single outlier can have a great impact on the mean).

- If the Data is **Strongly Skewed**. (+ve / -ve).

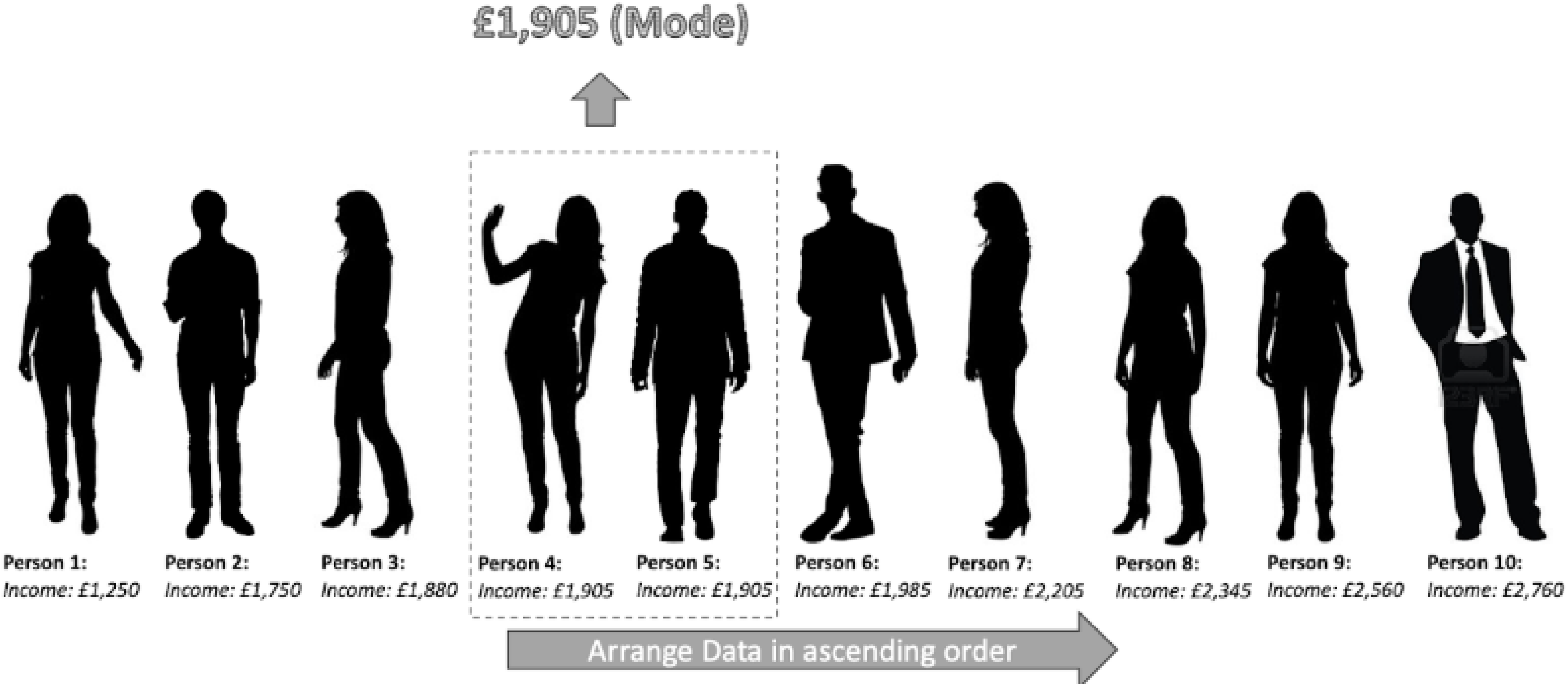
- If the given data is measured on **Ordinal Scale.** *(<u>ORDINAL</u> : Categorical Data with Relative Order / Ranking)*

  - Stage of cancer (stage I, II, III, IV)
  - Education level (elementary, secondary, college)
  - Pain level (mild, moderate, severe)

Measure of Central Tendency - MODE

The mode is the most common value.*

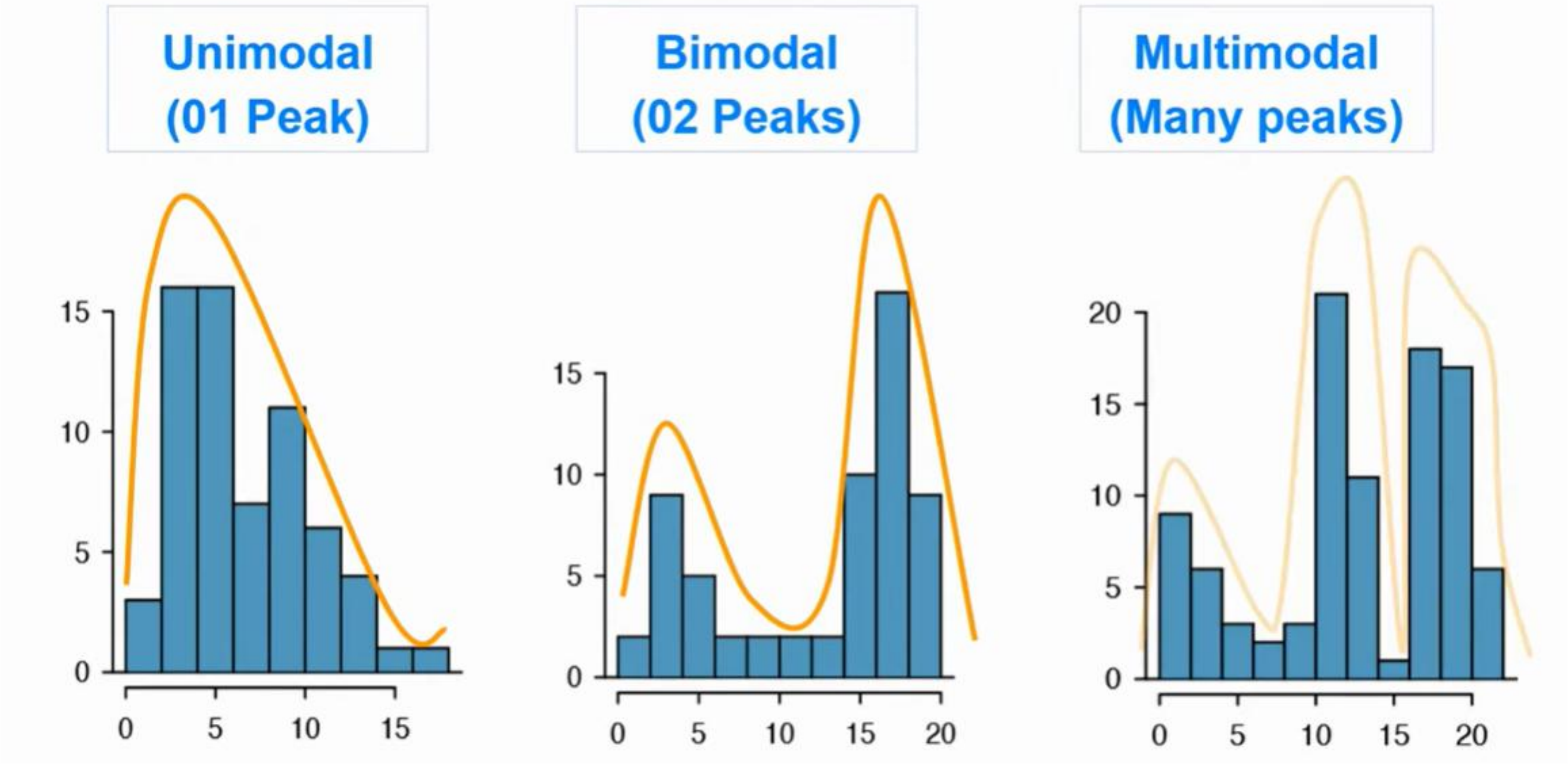
*Depending on your data you might have no mode ore multiple modes



Measure of Central Tendency - MODE

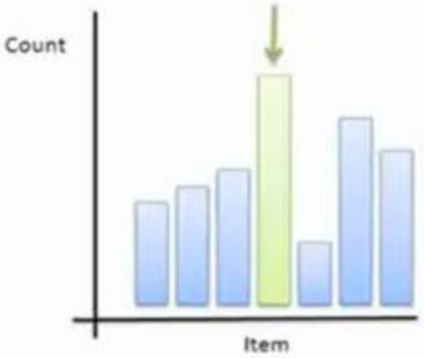
The value that **appears most frequently** in a dataset.

Types of mode: one mode(unimodal), two modes(Bimodal), Multimodal(many modes).



3. MODE

- Most Frequent Value in Data.
- Not susceptible to Outliers.



Examples ,

8, 10, 4, 6, 7, 5, 3, 21, 11, 98	→ No Mode
8, 4, 6, 7, 7, 7, 5, 3, 21, 10, 11, 10, 98	→ Mode : 7
8, 4, 6, 7, 7, 7, 5, 3, 21, 10, 11, 10, 10, 98	→ Modes : 7 ,10 → Two value with same frequency.

Two value with same frequency.

↑

Which mode describes the central tendency of the data ?

↑

That's Why Mode isn't always Suitable for Continuous data.

MODE , WHEN TO USE & WHEN NOT TO ?

For **Categorical** Data (mainly **Nominal Scale**).
Nominal : Categorical Data with NO Relative Order / Ranking).

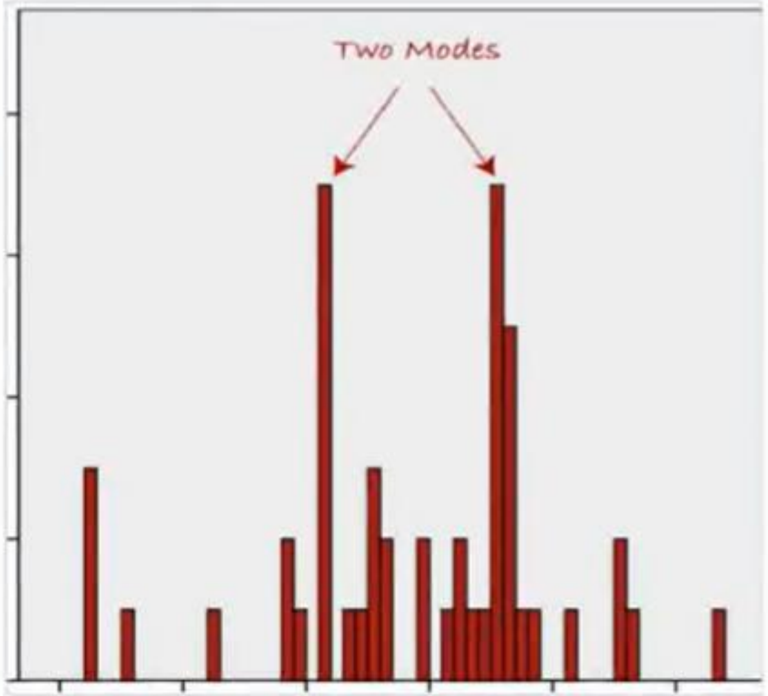
HAIR COLOR

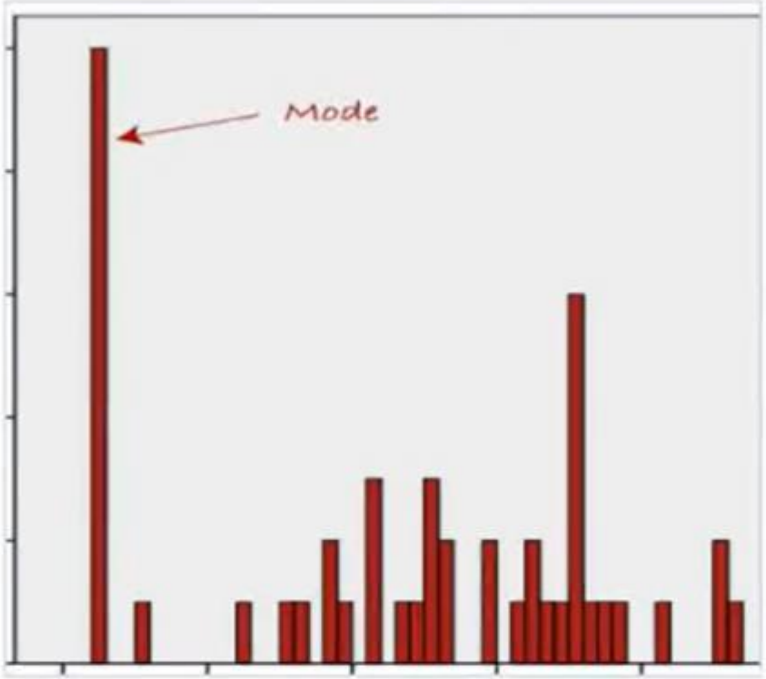
For **Continuous** data, **Not Always Suitable**

If Data is **Strongly Skewed**. (+ve / -ve).

Chances of getting 02 or more Modes.

Which mode describes the central tendency ?





LEARNING AUGMENTED

SUMMARY

To Find Best Measure of Central Tendency for a Given Dataset ...

- ❑ Check If **Variable** is **Categorical** or **Continuous** ?

- **If Variable is Categorical :**
  - ★ Check Measurement Scale - Nominal, Ordinal ?
    - → If **Nominal** : **MODE**
    - → If **Ordinal** : **MEDIAN**

- **If Variable is Numerical**
  - ★ For Both Measurement Scale : Interval , Ratio
    : *Check for Outliers & Skewness in Data*
    - → If **No Outliers** : **MEAN**
    - → If **Outliers** : **MEDIAN**
    - → If **No Skewness** : **MEAN**
    - → If **Skewed** : **MEDIAN**

IN SHORT

	Type of Variable	Best measure of central tendency
Categorical	Nominal	Mode
	Ordinal	Median
Numerical	Interval/Ratio (not skewed)	Mean
	Interval/Ratio (skewed)	Median

References

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